Palaeomagnetic Pole Positions: A Survey and Analysis

E. Irving*

Summary

The positions of the pole during different geological periods obtained from studies of the magnetism of rocks are reviewed. For the past 20 million years the results from different continents are in good agreement. For periods prior to this, the results from the same continent are still in broad agreement with one another but not with those from different continents. The meaning of this disagreement is discussed and it is concluded that explanations other than relative movements between continents (continental drift) are not promising.

1. Introduction

Most of our information about the Earth's magnetism has been obtained by direct observation over the past four hundred years. In principle, the study of the magnetism of rocks provides the means to extend this record back into the remote geological past since some rocks became permanently magnetized parallel to the Earth's field at about the time they were formed. Thus, by measuring these directions in the laboratory the behaviour of the field in the past can be investigated, but there is the disadvantage that the observations are indirect and there are many uncertainties. The reversals of magnetization in rocks indicative of periodic reversals in the field first attracted attention, but about four years ago palaeomagnetic results were used from the Tertiary by Hospers (1955) and from older periods by Creer & others (1954) to compute the position of the geomagnetic pole at these times, thereby fulfilling a prediction by Mercanton in 1926 that the study of rock magnetization could lead to investigations of polar movements. In recent years many workers have devoted their attention to defining the positions of the pole in this way and over one hundred results are now available from different countries. The principles of this aspect of palaeomagnetism are discussed in Sections 2 and 3, and an atlas of polar positions for different geological periods follows in Section 4. The significant features of the results are explained in Section 5, and in Section 6 there is a discussion of one of the most important of these, namely, the lack of agreement between results from different continents in pre-Miocene times.

2. Origin of the remanent magnetism in rocks

A rock which becomes permanently magnetized in the direction of the Earth's magnetic field at the time it is formed is said to have acquired a *primary* magnetization. Long after formation further components may be added modifying or obliterating the primary one and such rocks are said to possess a *secondary* magnetization. There are four principal ways in which a rock may acquire a permanent moment.

* Fellow in Geophysics, the Australian National University, Canberra, A.C.T., Australia.

Thermo-remanent magnetization. For any substance containing ferromagnetic minerals of uniform grain size and composition it is possible to define the relaxation time τ at a given temperature, τ being the time required for the magnetic moments to approach equilibrium with an applied field (Néel 1955). At temperatures very near the Curie point τ is very short, but it increases as the temperature falls and at normal temperatures becomes very long. Thus, an igneous rock which has cooled through its Curie temperature becomes magnetized parallel to the Earth's field and retains this direction of magnetization against subsequent changes in direction of the field provided τ exceeds its geological age. This is called thermo-remanent magnetization which is generally abbreviated to T.R.M.

Detrital magnetization. In sedimentary rocks the iron minerals are either detrital or chemical in origin and depositional processes or subsequent chemical activity may be responsible for the primary magnetization. Detrital ferromagnetic minerals are rock or crystal fragments usually of magnetite or specular ore which have been derived from pre-existing rock formations from which they may have inherited remanent magnetic moments. The conditions of deposition will affect the alignment of these particles as they become incorporated into the sediment, and in some cases, notably in the Torridonian Sandstone Series (Irving 1957) the dispersion of directions observed is much less in find-grained sandstones laid down under tranquil conditions than in coarse sandstones deposited in turbulent water (Figures 2 and 3), but the mean direction is the same. Sedimentary rock formations which have a predominantly detrital magnetization do not appear to be common, the best examples being the Quaternary varves (Johnson & others 1948, Ising 1942, Griffiths 1955, King 1955) and the Torridonian Sandstone Series. Buff, green and grey sediments which contain detrital iron minerals are usually found to have a weak secondary magnetization unrelated to the earth's field at the time of deposition (Nairn 1957b).

Chemical magnetization. Iron minerals formed by chemical activity in sedimentary rocks usually occur as iron oxide cement, or as coatings to the non-ferromagnetic detrital particles, or as fine iron oxide dust spread through the rock. At the time it is deposited it may become magnetized in the direction of the magnetic field (Martinez & Howell 1956, Doell 1957). Since these chemically deposited iron minerals are incorporated in the rock after deposition of the detrital non-magnetic mineral particles there is no dependence of dispersion on the grain size of the latter. The most important rock types of this category are the red beds usually associated with deposition in a hot desert environment, red and black haematitic limestones and red tuffs. Haigh (1958), who has demonstrated the efficacy of chemical magnetization in the laboratory, suggests that the mechanism by which chemical magnetization occurs is as follows. The relaxation times of minerals is low when their particle size is very small and such grains representing the initial phase of chemical deposition could become magnetized by the earth's field. As these small nuclei expand by crystal growth τ increases and a permanent magnetization not unlike T.R.M. is built up. The variable in this process is grain size, whereas in T.R.M. it is temperature.

Isothermal remanent magnetization (*I.R.M.*). For any magnetic mineral τ depends on grain size. Since the iron mineral assemblage in a rock specimen may contain a considerable range of particle sizes and chemical composition a wide range of

13652465b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

relaxation times may be represented so that the observed direction of the remanent moment will be the vector sum of primary and secondary components. Some success has attended efforts to remove the latter by "washing" specimens in alternating magnetic fields (Rimbert 1958, Creer 1958) or heating followed by cooling in zero field (Doell 1956, Cox 1957).

In general T.R.M. is responsible for the primary magnetization of igneous rocks. In sedimentary rocks the primary magnetization is detrital or chemical in origin. Between the period of formation and the present day the rock may become remagnetized by the process of I.R.M., or by T.R.M. if reheated above the Curie temperature, or by chemical magnetization if new minerals are deposited in the rock. Thus in thermally metamorphosed rocks the original magnetization will be replaced by a T.R.M. identifiable with the Earth's magnetic field at the time of metamorphosis. In rocks which have undergone metasomatic changes below the Curie point chemical magnetization is likely to be operative. This is of some interest since it should be possible to distinguish between granitized sediment and truly magmatic granite since the former is likely to have a chemical magnetization and the latter a T.R.M.

Figures 1 to 7 illustrate results from rocks which have acquired their magnetization in different ways. North-seeking directions are shown as dots or circles intersecting the lower and upper hemispheres respectively. Declinations are measured clockwise from geographic north. In Figures 1, 4, 5 and 7 the plane of projection is the present horizontal, which, since the rocks concerned are not tilted, corresponds to the horizontal at the time of deposition. In Figures 2, 3 and 6 the plane of projection is the bedding plane, so that the directions are corrected for folding. Figure 1 (Irving & Green 1957) shows the directions in a lava flow with T.R.M. Figures 2 and 3 illustrate detrital magnetization in sediments. The effect of chemical weathering is illustrated in Figures 4 and 5. The directions in unweathered pinkish-brown sandstone which is hard and compact are towards the south-west with shallow negative inclinations, whereas those in weathered sandstones from the *same* formation are clustered around the present axial dipole field. The weathered rock is bright red, friable and highly porous, and has undergone chemical weathering in some cases to a depth of several hundred feet during the Cenozoic. The iron minerals present in the fresh sandstone have been reconstituted and there has been a good deal of iron enrichment. A further case of chemical magnetization is illustrated in Figure 6 which shows the directions in some Carboniferous red tuffs. Their colour is due to haematite dust which is responsible for the magnetization and which was probably excreted as a by-product of devitrification of volcanic glass (Coombs 1958). Lastly, Figure 7 illustrates the effect of a secondary isothermal magnetization superimposed on a primary magnetization (Creer 1957b). The directions fall in the plane containing the primary direction and the present dipole field.

3. Geomagnetism and palaeomagnetism

Mathematical analysis ascribes the Earth's field to hypothetical multipoles at the geocentre (Vestine & others 1947). The lowest order multipole, or dipole, has three components, one along the axis called the geocentric axial dipole which is by far the largest, and two of much smaller value in the equatorial plane. The higher-order multipoles are small in magnitude compared with the geocentric axial dipole. Changes affecting the geomagnetic field are of four main types each with characteristic periodicities: the transient and secular variations which have

13652465b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

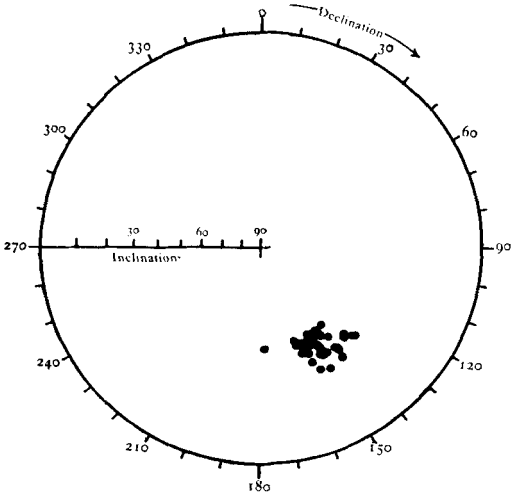


FIG. 1.—*Directions of magnetization in a Cenozoic lava flow.* Directions in samples taken from a quarry (Adam's Quarry, Alphington, Melbourne, Australia) in the Newer Volcanics of Victoria. The stereographic projection is used in this and in Figures 2 to 17.

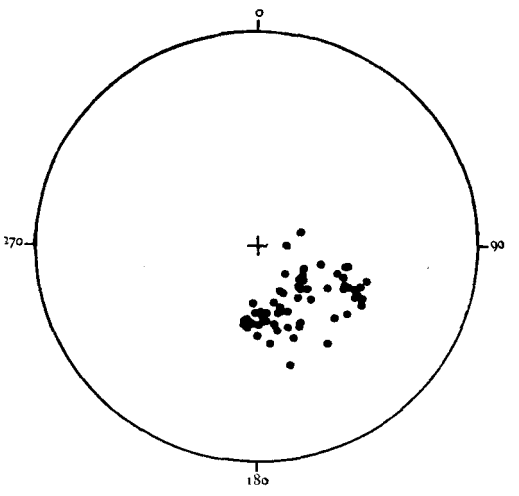


FIG. 2.—*Directions in very fine sandstones.* These specimens were collected from the Aultbea Group of the Torridonian Sandstone Series, Scotland. The specimens are spaced through a section 75 feet thick on the shore near Laide, Rosshire. The age is Upper Pre-Cambrian.

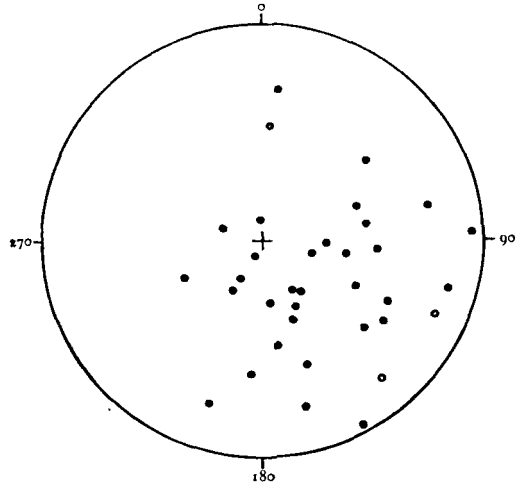


FIG. 3.—*Directions in coarse-grained current-bedded sandstones.* Results from the Aultbea Group of the Torridonian sandstone on Isle of Ewe, Rosshire. These specimens are from the base of the Group several hundred feet beneath the beds from which the results in Figure 2 were obtained. The directions are more scattered but the mean direction is the same as in Figure 2.

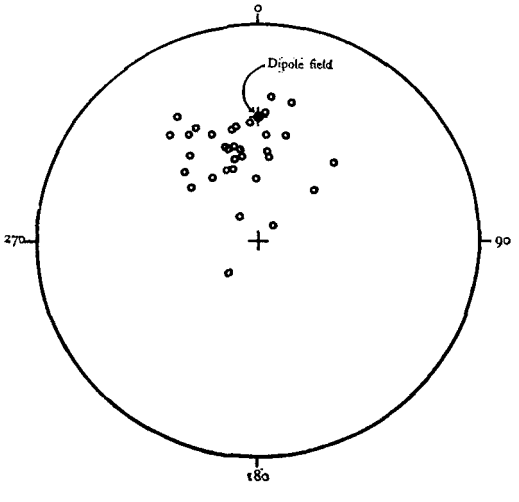


FIG. 4.—*Directions of magnetization from weathered Cambrian sandstone.* Elder Mountain Sandstone, northern Western Australia.

13652465b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

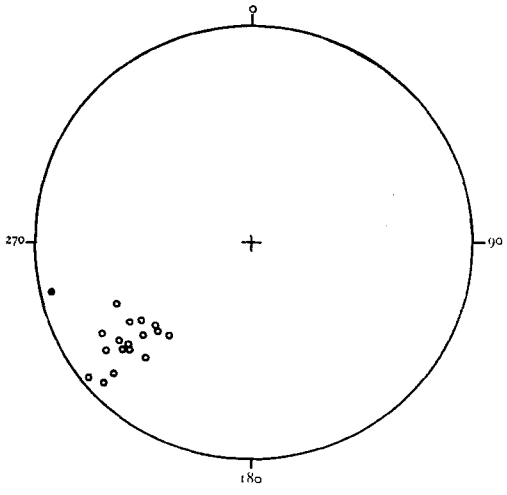


FIG. 5.—*Directions of magnetization in fresh Cambrian sandstone.* Elder Mountain Sandstone, northern Western Australia.

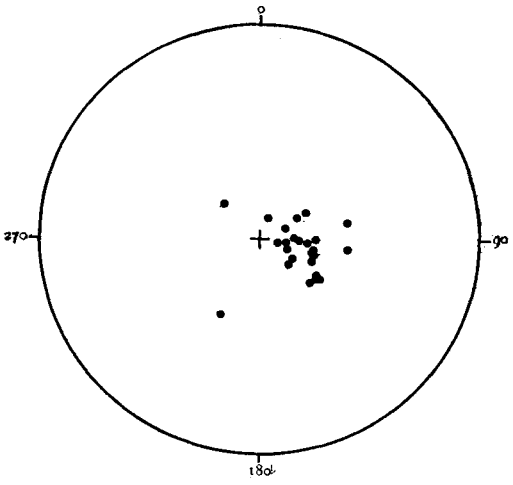


FIG. 6.—*Directions of magnetization in fine-grained red tuff.* Eagleton quarry, Seaham, New South Wales. Kuttung Series, Upper Carboniferous.

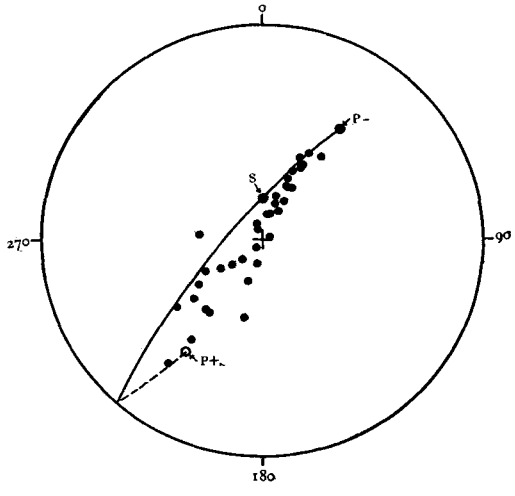


FIG. 7.—*Directions of magnetization in Triassic sandstones.* Sidmouth, England (after Creer 1957b). P^- and P^+ are the poles of the axis of primary magnetization obtained by Clegg & others (1954). The directions fall close to the plane containing this and the direction of the axial dipole field *S*.

13652465b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

their major effect on the non-dipole components, and reversals of polarity and change in attitude of the dipole field. The evidence for the first of these comes from direct observation and for the third and fourth from rock magnetism. Information on secular variation is derived from both sources but chiefly from direct observation.

The transient variations have periods of the order of days and have negligible effect on the direction of the field. The transient variations arise primarily from magnetic disturbances in the ionosphere (Chapman & Bartels 1951) whereas the remainder of the geomagnetic field is produced internally probably by electrical currents driven by fluid motions in the Earth's core (Elsasser 1956, Runcorn 1954, Bullard & Gellman 1954, Hide 1956).

The secular changes have angular amplitudes of about 10–20° and periods of the order of 10^2 or 10^3 years. The secular variation is represented by maps showing isopors or lines of equal change. Maps for the last few decades show a westerly drift of the isopors at a rate of 0°·3 of longitude per year (Bullard & others 1950). Thus, the field when viewed from the Earth's surface for 2 000 years or more should average to the geocentric axial dipole. This has been substantiated by palaeomagnetic studies (Figure 8) and has been the most important feature of the field at least during the past 10^6 years. If this has always been so the average direction of magnetization of a rock sequence which took several thousands of years or more to be formed will approximate to a line of force in the average dipole field, and, the poles or intersection of the dipole axis with the surface can be readily calculated and errors ascribed (Creer & others 1957).

In early work when averages through thick rock formations were taken (Creer & others 1954) polar wandering was represented as a steady drift of the dipole axis at the rate of 0°·3 of arc per 10^6 years, so that in the late Palaeozoic the pole was approximately 60° from its present position. This may be the smoothed effect of many small irregular polar jumps, and Green (1958) has shown that the polar data from Europe and the United States may equally well be represented as a random walk on a sphere with instantaneous velocities of 3° per 10^6 years, ten times the smoothed velocity. Thus, to obtain information about polar wandering it is necessary to eliminate the secular variation by averaging through a rock formation which was deposited over a period exceeding 10^3 years but if the period covered exceeds 10^6 years it is likely that some of the details of polar wandering will be lost.

Reversals of polarity occur within many rock formations (see Table 1 column 5). These could arise from self-reversal of the constituent ferro-magnetic minerals (Néel 1951, Nagata 1953), or from reversals of the Earth's field implying periodic reversals in polarity of the geocentric dipole (Hospers 1954, Allan 1958 and many others). Reversals appear to be sudden events, geologically speaking, being accomplished in about 10^4 years. They seem to have occurred about once every 10^6 years since the middle of the Tertiary. The occurrence of reversals creates an ambiguity as to the sign of the dipole axis.

The computation of pole positions has as its basis two assumptions; that in certain rocks the directions of magnetization reflect the ancient field direction, and that the field has averaged to a geocentric dipole. These assumptions, which are discussed in Section 6, are concerned with events long since past and by their nature can never be proved absolutely true or false. They have the merit that they lead to results which are internally consistent (see Section 6) and which are of some interest for geophysics and geology.

13652463b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

4. Palaeomagnetic pole positions

The pole positions obtained by this method are now reviewed in geological sequence. Numerical data are given in the table. In some cases the pole positions were not calculated by the workers who made the observations, and this has been done by the author. These cases are indicated by an asterisk * in the sixth column. Many results have been obtained from rock formations of substantial thickness so it can be reasonably supposed that the secular variation has been averaged out. Errors for these poles have usually been given, but there are cases in which the necessary numerical data has not been included in the original and these are indicated by question marks in the last two columns. Many results have been obtained from limited thicknesses, perhaps a single lava flow or a few feet of sediment. These are spot readings in time and will be subject to an unknown error due to secular variation and are indicated by dashes in columns 9 and 10. They are included for comparison with results from thicker sequences. Because of the drift of the dipole axis it is difficult to describe the polarity of the field in any consistent fashion. Runcorn (1955) suggests that dipole fields with the same sense as the present field relative to the axis of rotation be termed negative and opposed fields positive. This is satisfactory for Carboniferous and later pole results which lie in more or less connected sequences terminating at the present rotation poles. The distribution of earlier poles is however much more complicated and their identification with one or other of the geographic poles is arbitrary. The convention used here is to regard the field as negative (positive) when the pole towards which the north (south) seeking magnetization points is in the present day northern hemisphere. These polarities are recorded in the fifth column of the table. The poles for the northern hemisphere are plotted in Figures 8 to 17. The mean position is indicated by a dot and where possible is circumscribed by an oval within which the pole lay at $P = 0{\cdot}95$ (Creer & others 1957). The poles are identified by the numbers **1** to **112** always in heavy type, the numbering being consistent in the Figures and Table.

Quaternary (Figures 8 and 9). In some cases the rock formations studied contain beds of both Pliocene and Pleistocene age and for this reason it is convenient to include some Pliocene results here. In Europe and North America results are available from rock formations laid down in post-glacial and historic epochs. The directions of magnetization in 11 lava flows of Mt Etna covering a period of 2305 years have been measured by Chevallier (1925) (pole **1**). Two sections of varved sediments spread over a period of 1850 years have been studied by Griffiths (1955) samples being taken approximately every 20 years (**3**). Pole **4** has been obtained from the mean direction of 8 post-glacial lava flows (Hospers 1955) from Iceland covering a period of about 5000 years, a result which has been recently confirmed by Brynjolfson (1957). A sequence of varves from New England covering a period of 6000 years have been studied by Johnson & others (1948) (**10**).

Results from the Pleistocene and Late Pliocene have been reported from many parts of the world. Results from 8 lava flows in Central France which straddle the Plio-Pleistocene boundary have been recorded by Roche (1951) (**2**). Hospers (1955) has given results from the Early Quaternary lava flows of Iceland (**5**). This work has been extended by Sigurgeirsson (1957) who has also studied Pliocene lava flows. Sigurgeirsson considers flows with negative and positive magnetization separately (**6** and **7**) and does not distinguish between Pleistocene and Pliocene flows. In Japan nine volcanic groups have been studied from Central Honshu

13652465b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1

Pole positions obtained from palaeomagnetism. Northern hemisphere polar co-ordinates are given; $d\psi$ and $d\chi$ are the semi-axes of the elliptical error area round the pole at a probability of 95 per cent, $d\psi$ in the colatitude direction and $d\chi$ perpendicular to it. Other symbols are explained in section 4.

	Country	Rock Units	Age	Sign	Pole Positions				
					No.	Lat. N	Long.	$d\psi$	$d\chi$
Quaternary	Italy	Lava flows of Mt. Etna	394 BC to 1911 AD	−	*1	86·3	125·5E	7·3	9·9
	France	Lava flows of Central France	Villafranchian and Pleistocene	+	2	77·9	93·3E	15·7	20·3
	Sweden	Varved sediments of the Angerman River	1100 BC to 750 AD	−	*3	88·4	160·2E	12·0	13·0
	Iceland	Post glacial lava flows of S.W. Iceland	Approximately 5000 BC to 0 AD	−	*4	86·0	151·9E	13·3	14·8
		Lava flows of W. and S.W. Iceland	Early Pleistocene	+	*5	87·6	146·8E	11·5	12·6
		Normal lava flows	Pliocene and Pleistocene	−	6	87·8	149·0E	6·7	6·7
		Reversed lava flows	Pliocene and Pleistocene	+	7	77·3	73·9E	6·7	6·7
	Japan	North Izu and Hakone volcanic regions	Lower Pleistocene through to Holocene	+ & −	*8	76·4	37·6E	7·1	10·4
		Kawajiri basalt	Early Pleistocene	+	*9	80	49E	—	—
	U.S.A.	Varves of New England	13000 BC to 7000 BC	−	*10	80·5	131·8E	9·5	13·6
	Argentina	Lava flows	Quaternary	+ & −	11	83	126E	6	7
	Australia	Newer Volcanics of Victoria	Pliocene, Pleistocene and recent	+ & −	12	86·3	77·9W	5·5	7·2
Neogene	France	Lavas of Mt Dore and the Limagne	Pontian and Pliacentian	+	*13	73·1	167·3W	13·6	19·2
	Iceland	Lavas of the north and north-west	Miocene	+ & −	*14	88·6	5·2E	9·6	10·3
	U.S.A.	Columbia River basalts	Miocene and Pliocene	+ & −	*15	87	40E	4·5	3·6
	New Zealand	Ignimbrites of the North Island	Pliocene	—	*16	79·2	35·2E	?	?

13652464b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Palaeogene and Cretaceous	Scotland	Mull lavas	Lower Tertiary, probably Eocene	+	*17	82	71W	28	31
		Mull Intrusives	Lower Tertiary	+ & −	*18	72	139E	11	14
		Dykes of Arran		+ & −	*19	79	154E	12	15
	England	N.W. dykes of Northern England	Oligocene	+	*20	86	76W	21	25
	Ireland	Lower basalts of Antrim	Lower Tertiary, probably Eocene	+	*21	74	133E	6	8
	France	Intrusive basalt of the Limagne	Aquitainian	+	*22	77	3E	—	—
		Sills and dykes of Gergovie		+	*23	73	119E	12	16
	India (Deccan Traps)	Lavas from Bombay State	Late Cretaceous to Eocene	+ & −	24	28	78W	10	15
		Linga area, lavas near base of the sequence		+	25	37	83W	13	19
		Khandala area, upper part of middle of sequence		+	26	25	79W	14	18
		Kambatki ghat		+	*27	31	103W	—	—
		Igatpuri ghat		+	*28	36	87W	—	—
		Nipani area, lower flows		+	29	35	95W	—	—
		Nipani area, upper flows		−	30	52	70W	3	5
		Amba ghat, lower flows		+	31	27	76W	8	10
		Amba ghat, upper flows		−	32	52	64W	—	—
		Mt Pavagadh, lower flows		−	33	58	91W	4	8
		Mt Pavagadh area, upper tuffs	Lower Tertiary	−	34	75	89W	—	—
	U.S.A.	Siletz volcanics	Lower and early middle Eocene	+ & −	35	37	50W	7	10
		Laney shale, Wyoming	Eocene	−	*36	86	164W	8	10
	Australia	Older Volcanics of Victoria	Lower Tertiary, probably Eocene	+ & −	37	66·8	57·3W	10·8	12·1
	U.S.A.	Dakota Sst.	Cretaceous	+	38	77	127E	9	11
Jurassic	Scotland	Lsts. and ssts. from Skye and Brora	Jurassic	+ & −	39	56	76E	—	—
	India	Rahajmahal traps	Jurassic	−	40	13	70W	7	8
	Rhodesia	Karroo basalt	Jurassic	−	41	60	77W	4	6
	South Africa	Karroo dolerites (surface samples)	Jurassic	+ & −	42	76	128W	14	19
		Karroo dolerites and baked sediments (Underground samples)	Jurassic	+ & −	*43	69	93W	13	13
	Brazil	Serra Geral lavas and baked Botacatu sst.	Jurassic	+ & −	44	83	126W	2	3
	Australia	Tasmanian dolerites	Jurassic	+ & −	45	50	23W	7	7

13652464b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1—*continued*

	Country	Rock Units	Age	Sign	Pole Positions				
					No.	Lat. N	Long.	$d\psi$	$d\chi$
Triassic	England	New Red Sandstone	Triassic	+ & −	*46	43	131E	7	12
	France	Red sst. of the Vosges	Triassic	+	*47	28	143E	6	12
	U.S.A.	Lavas and sediments Connecticut valley		+ & −	48	55	88E	8	15
		Lavas Connecticut valley	Upper Triassic	−	49	54	90E	6	11
		Brunswickian formation of Newark Series	Upper Triassic	−	50	63	93E	3	6
		Springdale sst.		−	51	55	107E	—	—
	Bechuanaland	Kgoma Series	Upper Triassic	−	52	54	44W	5	10
	Australia	Brisbane tuff.	Triassic	+ & −	53	39	37W	—	—
Permian	Scotland	Mauchline lavas	Permian, most likely belong to the lower Permian	+	54	36	175E	3	8
		Mauchline sediments	Permian, most likely belong to the lower Permian	+	55	37	163E	6	12
	England	Exeter Traps	Permian, most likely belong to the lower Permian	+	56	43	164E	10	20
	France	Esterel lavas (flow R3)	Permian, most likely belong to the lower Permian	+	57	45	131E	—	—
		Esterel lavas (flow R4)	Permian, most likely belong to the lower Permian	+	*58	46	141E	—	—
		Esterel dolerite	Permian, most likely belong to the lower Permian	+	*59	53	165W	—	—
		Nideck porphyry	Permian (Saxonian?)	+	60	43	168E	2	5
		Montcenis sst.	Saxonian	+	61	38	162E	—	—
	Norway	Lavas of Oslo graben	Permian	+	62	46	165E	7	11
	Germany	St. Wendel sst.	Permian	+	63	45	175W	—	—
	U.S.A.	Supai beds (Runcorn)	Lower Permian	+	64	26	121E	6	8
		Supai beds (Doell)	Lower Permian	+	65	39	115E	4	7
		Supai beds (Graham)	Lower Permian	+	*66	41	117E	—	—
	Kenya	Majiya Chumvi	Upper Permian	+	67	4	150E	—	—
	Kenya	Taru grit	Lower Permian	?	68	0	87E	—	—
	Australia	Lower Marine Series	Lower Permian	+	69	38	6W	—	—
		Upper Marine Series	Middle Permian	+	70	27	11W	21	21

13652464b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Carboniferous	England	Pennant sst.	Upper Carboniferous	−	*71	43	114E	—	—
		Clee Hills sills and contact rocks		+	*72	27	155E	?	?
		Tideswelldale baked rocks		+	*73	5	143E	?	?
		Lundy Granite		+	*74	43	180E	—	—
	U.S.A.	Naco Sst.		+	75	41	120E	4	8
	Australia	Kuttung (sediments)		+ & −	76	32	15W	12	12
		Kuttung (lavas)		−	77	43	30W	16	16
	England	Toadstones	Lower Carboniferous	−	78	55	148E	11	17
	Scotland	Kinghorn flows (41–46)		−	*79	8	153E	?	?
		Kinghorn flows (48–54)		+	*80	12	158E	?	?
		Kinghorn flows (64–65)		−	*81	42	150E	—	—
	U.S.A.	Barnett formation (normal)		−	82	42	142E	—	—
		Barnett (reversed)		+	83	41	128E	?	?
	England	Shatterford intrusion	Carboniferous undifferentiated	+ & −	*84	32	137E	—	—
		Sediments from the northern counties		−	*85	51	143E	?	?
Devonian	Britain	Old Red Sandstone	Lower and Upper Devonian	+	86	30	159E	3	5
		Brownstones	Lower Devonian	+	87	25	102E	—	—
	Australia	Ainslie volcanics	Lower Devonian?	−	88	66	168W	7	13
Silurian	U.S.A.	Clinton iron ore	Middle Silurian	+	89	35	138E	—	—
		Rosehill fm.	Middle Silurian	−	*90	19	138E	4	6
	Australia	Mugga porphyry	Upper Silurian	−	91	60	157W	—	—
Cambrian	Wales	Caerbwby sst.	Lower Cambrian	+	92	15	173E	8	10
	U.S.A.	Wilberns fm., Point Peak member	Cambrian	?	93	0	158E	—	—
	Australia	Elder sst.	Middle Cambrian	+	94	34	172W	5	10
		Antrim plateau basalts	Lower Cambrian, possibly uppermost Proterozoic	+ & −	95	36	154W	6	12

13652464b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1—*continued*

	Country	Rock Units	Age	Sign	Pole Positions				
					No.	Lat. N	Long.	$d\psi$	$d\chi$
Pre-Cambrian	England	Longmyndian	Upper Pre-Cambrian	+ & −	96	2	120W	7	13
	Scotland	Upper Torridonian Sandstone Series		+ & −	97	6	43E	4	6
		Lower Torridonian Sandstone Series		−	98	35	118W	5	8
	U.S.A.	Portage Lake Group, lavas	Middle Keweenawan	−	99	25	170W	5	7
		Copper Harbour Group, red sst. and lavas	Upper Keweenawan	−	100	30	176E	5	8
		Freda and Nonesuch Group, red sst. and shl.		−	101	9	169E	2	3
		Jacobsville Group, red sst.		−	102	14	10E	7	13
		Chequamegon Group, sandstones		−	103	69	47W	10	11
		Michigan diabase dykes	Upper Pre-Cambrian	+	104	45	99W	2	2
		Hakatai shales (Runcorn)		−	105	31	150W	9	10
		Hakatai shales (Doell).		−	106	18	144W	27	30
		Hazel fm., flat lying red beds	Pre-Cambrian	−	107	49	175W	?	?
		Hazel fm., folded red beds		−	108	60	151E	?	?
	Africa	Pilansberg dykes	$1290 \pm 180 \times 10^6$ years	−	109	8	43E	?	?
	Australia	Buldiva Quartzite	Upper Pre-Cambrian	+	110	30	121W	8	14
		Nullagine lavas		+	111	51	18W	10	13
		Edith River Volcanics		+	112	6	14W	15	24

13652464b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

(Nagata & others 1957) and pole **8** has been obtained from an average of results from all these groups. The mean direction in one flow from western Honshu has been reported by Asami (1954) and pole **9** calculated from this is in good agreement with the results from Central Honshu. Much detailed work on sediments in Japan has been done but numerical values are not given and pole positions cannot be determined. In the southern hemisphere results are available from some Quaternary volcanics in Central Argentina (Creer 1958) (**11**) and from the Newer Volcanics of the western district of Victoria (**12**) (Irving & Green 1957). Thirty-two sites have been sampled at the latter which are for the most part of Pleistocene age although some of the younger flows are Recent and some of the oldest flows may be Pliocene.

All the post-glacial poles (Figure 8), and, with two exceptions, all the Plio-Pleistocene poles (Figure 9) coincide within their errors with the geographic pole.

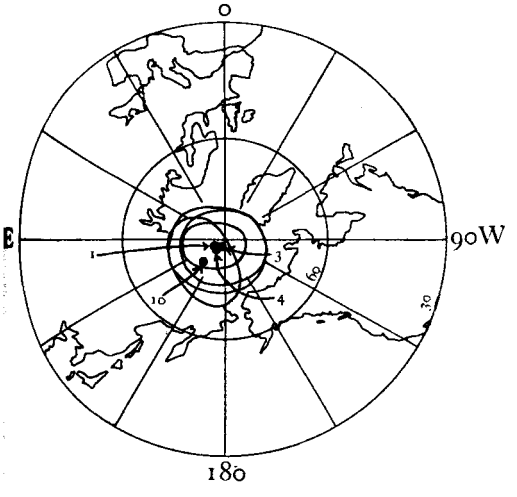


FIG. 8.—*Pole positions for post-glacial times.* The poles may be identified by referring to the Table.

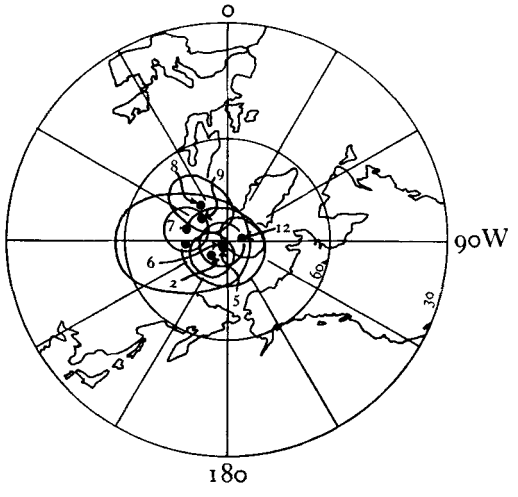


FIG. 9.—*Pole positions for Pleistocene and Plio-Pleistocene times.*

The divergence in pole **7** derived from the Plio-Pleistocene flows of Iceland is curious since these flows which have positive polarity are interbedded with flows with negative polarity whose directions of magnetization average to the dipole field (**6**). Secondary components of magnetization could be responsible but care has been taken to eliminate these by "washing" the specimens in alternating fields (Sigurgeirsson 1957). Since the sample size is fairly small (29 specimens) this result needs confirmation or rejection by further measurements. In Japan the declinations are consistently westerly and in view of the large number of specimens measured spanning the period from early Pleistocene to Holocene this divergence is probably significant. It could be explained by an anti-clockwise rotation of Japan by $16° \pm 8°$ during the past million years.

Neogene. (Figure 10). Roche (1951) has obtained the directions of magnetization from 2 lava flows of Upper Miocene and 3 of Lower Pliocene age (pole **13**). Hospers (1954) sampled over 100 flows in the Neogene plateau basalts of Iceland (**14**). Seventy-three flows in the Columbia River basalts have been sampled by

13652465b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Campbell & Runcorn (1956) (**15**). Pole **16** is from the ignimbrites of New Zealand (Hatherton 1953). Poles for the Iceland lavas and the Columbia River basalts which are the most thoroughly sampled formations coincide, within the errors, with the geographic pole. The small divergence of the French pole (3° at the 95 per cent confidence limits) could be due to the comparatively small number of flows sampled.

Palaeogene and Cretaceous (Figure 11). The lava flows of Mull and certain dykes have scattered directions of magnetization in some cases random (Bruckshaw & Vincenz 1954, Vincenz 1954). Directions in 8 flows and 5 dykes are not random and poles **17** and **18** have been calculated from these. The dykes of Arran and their contact sediments (Leng 1955, 10 sites) and the north-west trending dykes of

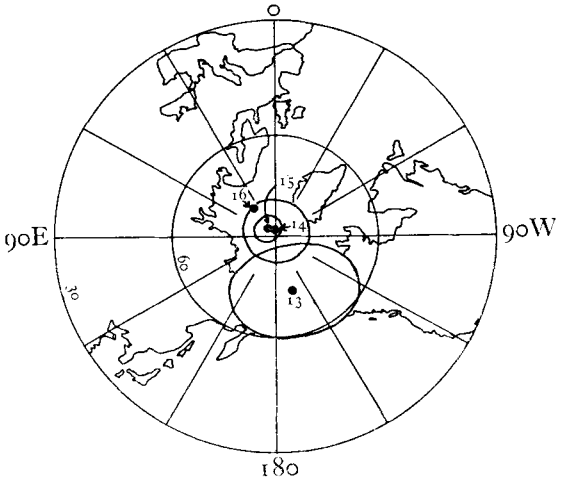


FIG. 10.—*Neogene pole positions.*

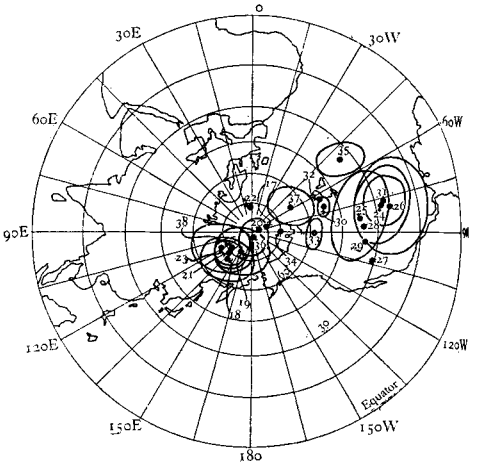


FIG. 11.—*Palaeogene and Cretaceous pole positions.*

northern England (Bruckshaw & Robertson 1949, 7 sites), which are representative of the last phase of Tertiary igneous activity in Britain, give the poles **19** and **20**. Twenty-four flows from the Lower Basalts of Northern Ireland have been sampled by Hospers & Charlesworth (1954) (**21**). The directions of magnetization of the intrusives in the Limagne and Gergovie (**22**, **23**) have been surveyed by Roche (1950 a and b, see Hospers 1955). The Mull lavas and N.W. dykes of England have very large associated errors which are not plotted on Figure 11 to avoid confusion. The other European poles are better defined and there is a tendency to occupy easterly longitudes in latitudes between 70 and 80°.

Many results are available from the Deccan Traps of India (Irving 1956a, Clegg & others 1956, 1957, Deutsch & others 1958). The traps are of uppermost Cretaceous or Eocene age with the possibility that some of the highest levels may be as young as Oligocene (discussion in Krishnan 1949). Samples have been taken from many places but unfortunately the stratigraphic relation of these localities to one another is generally not known. Samples at 9 sites in the Linga area (**25**), 20 sites covering 2 050 feet in the Khandala area (**26**), 5 sites spaced through 1 300 feet in the lower part of Amba Ghat (**31**), 7 sites in Bombay State (**24**), 3 lava flows at Igatpuri Ghat (**28**), a single flow at Kambatki Ghat (**27**), and one

13652465b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

flow in the Nipani area (**29**) give pole positions in the region of the southern United States. This represents good agreement from many determinations. Results (**32**, **30**) from the upper flows at Amba (25 feet) and at Nipani (5 sites), from the lower flows (8 flows pole **33**) and upper tuffs (2 sites pole **34**) at Mt Pavagadh give poles in similar longitude but more northerly latitudes. The former groups have positive polarization whereas the latter are all negative. The differing latitudes of the poles could therefore be due to secondary I.R.M. However, at any one locality the upper sections always give poles nearer to the geographic pole than lower sections and could be due to polar wandering during the time the Traps were being laid down. An alternative proposal is that differing depths of burial have produced these differences.

Apart from work by Cox (1957) (**35**) on the Siletz volcanics of Oregon few results are available from the Palaeogene of North America. Early work by Torreson & others (1949) on Tertiary sediments in the Western United States suffered from a lack of adequate stability evidence (Hospers 1955). However, for comparison with Cox's work a result (**36**) from the Laney shale is given. This pole coincides with the present geographic pole whereas Cox's pole is in mid-Atlantic. Cox has studied 8 flows spread through about 4000 feet, and reversals occur. It is somewhat unlikely therefore that its divergence from the present pole is due to failure to average out the secular variation. It is noteworthy that the distance of pole **35** from Oregon, $55° \pm 7$, is not greatly different from the present co-latitude of 45° and the large observed divergence could, in part at least, have resulted from a clockwise rotation of this particular part of Oregon since the middle Eocene. The region has undergone orogenic movement since that time so this possibility has to be borne in mind. Measurements on undisturbed Eocene rocks from other regions of North America should settle the matter.

Runcorn (1955) has given the pole from North America for the Cretaceous Dakota Sandstones (**38**). Certain Turonian (Upper Cretaceous basalts from Madagascar have directions of magnetization close to the present field (Roche & others 1958) and could possess a secondary magnetization. Pole **37** has been obtained from samples from 15 sites in the Older Volcanics of Victoria, Australia (Irving & Green 1957).

Jurassic (Figure 12). Clegg & others (1958) have given a result (**40**) from the Rahajmahal traps of India. Eight lava flows from Bulawayo and the Victoria Falls sampled by Nairn (1956) give pole **41**. In the Union of South Africa samples have been collected (Graham & Hales 1957) from 20 sills and dykes (pole **42**) at surface exposures, and from sills and associated baked sediments in two mines. The underground samples are from 2 sills at Winkelhaak in the Transvaal and 2 sills and baked sediments at Estcourt in Natal and pole **43** has been calculated from a mean of these, thereby probably eliminating secular variation. Creer (1958) has given the pole (**44**) for the Serra Geral volcanics and associated sediments in Southern Brazil, and Irving (1956b) the pole **45** from 30 sites in the Tasmanian dolerites. The formations referred to above are usually regarded as Jurassic, but in some cases, for instance, the Tasmanian dolerites, the possibility of a Cretaceous age although much less probable cannot be excluded. From the northern hemisphere there is only one preliminary result (**39**) obtained from various sedimentary rocks in Scotland (Nairn 1957b).

Triassic (Figure 13). Clegg & others (1954, 1957) have obtained results from the Triassic of England and France, nine sites in each case (poles **46**, **47**). There are

13652465b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

four determinations (**48**, **49**, **50**, **51**) from the Upper Triassic of the United States (Du Bois & others 1957). The determinations from South Africa (**52**) and Australia (**53**) have been given by Nairn (1957a) and Irving & Green (1958).

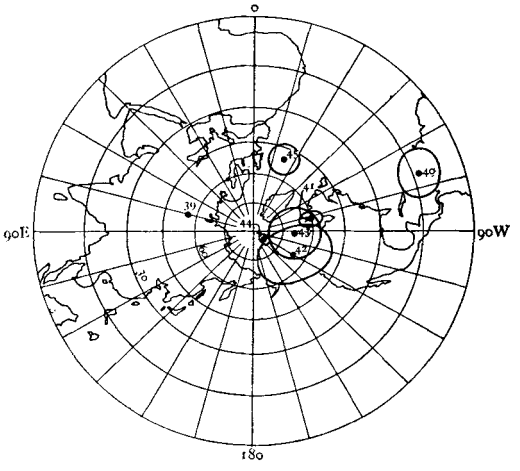


FIG. 12.—*Jurassic pole positions.*

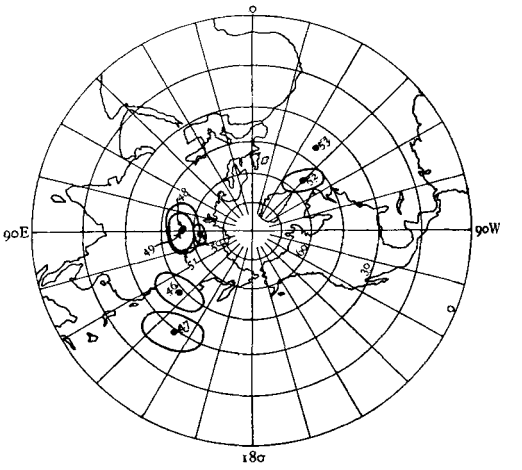


FIG. 13.—*Triassic pole positions.*

Permian (Figure 14). Results from samples spaced through 500 feet of basalt at Mauchline in Scotland give pole **54**, and **55** is obtained from sampling through 2 100 feet of the overlying red sandstones (Du Bois, 1957). Pole **56** has been obtained by Creer (1957a) from 5 lava flows in Devonshire. Determinations from two flows and one dolerite in the Esterel lavas of Central France (Roche 1957,

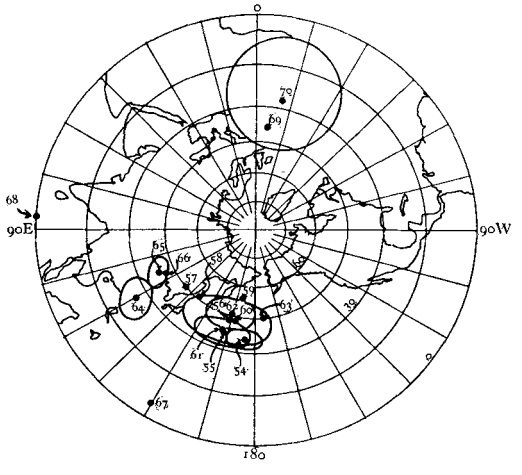


FIG. 14.—*Permian pole positions.*

Rutten & others 1957) give poles **57**, **58** and **59**. The porphyries of Nideck, the red sandstones of St Wendel in Saarfalz, and the red sandstones of Montcenis have provided poles **60**, **63** and **61** (Nairn 1958). Rutten & others (1957) have obtained results from 15 lava flows near Oslo (**62**). There are three independent determinations (**64**, **65**, **66**) from the Supai beds in Arizona (Runcorn 1955, Doell 1955, Graham 1955). Poles have also been given by Nairn from Kenya (**67**, **68**) (Creer

13652463b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

& others 1958), and from Australia there are two results (Irving & Green 1958), **69** and **70**, the former being based on the one lava flow from the Hunter River Valley and the latter from 3 flows from the Illawara coast.

Carboniferous (Figure 15). Carboniferous rocks in Britain have been extensively studied. Results from a single site in the Pennant sandstone (**71**) have been reported by Clegg & others (1954). Clegg & others (1957) have given results from

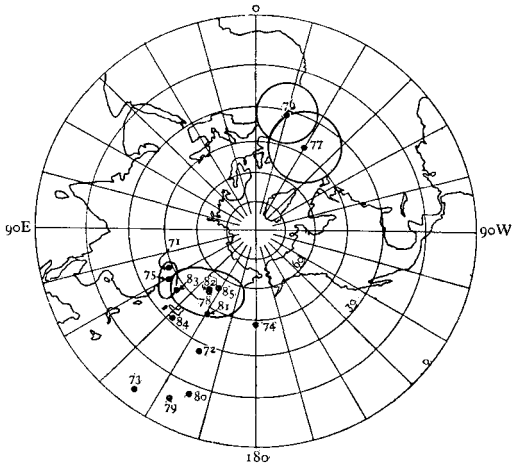


FIG. 15.—*Carboniferous pole positions.*

igneous rocks and associated baked sediments at Clee Hill (6 sites pole **72**), Tideswelldale and Shatterford (2 sites). They have also studied 3 groups of 5, 6, and 2 lava flows from Kinghorn (**79, 80, 81**). Belshé (1957) has given results from igneous and sedimentary rocks (**78, 85**). A result by Blundell (1957) from Armorican granites of Lundy Island gives pole **74**. The poles from Britain are scattered widely over the north-eastern Pacific, but unfortunately the errors in this work have not always been published and in many cases (**71, 81, 84**) limited thicknesses have been sampled so that it is impossible to assess the errors due to experiment and secular variation. Runcorn (1955) has sampled from 200 feet of Naco sandstone (**75**). Howell & Martinez (1957) have sampled from one site (**82**) and 8 sites (**83**), from the Barnett formation. From Australia there are results from sedimentary (**76**) and igneous rocks (**77**) in the Kuttung Series of the Hunter Valley.

Devonian (Figure 16). Creer (1957a) has given results from 35 sites in the Lower and Upper Old Red Sandstone of the Welsh cuvette (**86**). Results from one locality in the Brownstones (Clegg & others 1954) give pole **87**. Results from igneous rocks at Mt Ainslie near Canberra give pole **88** (Irving & Green 1958); there is some uncertainty as to whether these beds are Silurian or Devonian.

Silurian (Figure 16). Seven samples from a limited vertical sequence in the Clinton iron ore gives pole **89** (Howell & others 1958). Samples from 5 sites in the somewhat older Rosehill formation (Graham 1949) give pole **90**. No results are available from the Ordovician. Pole **91** is from a porphyry intrusion near Canberra (Irving & Green 1958).

13652465b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Cambrian (Figure 16). Cambrian results are available from the Caerbwby sandstone in Wales (**92**) (Creer 1957a), the Point Peak member of the Wilbern's formation (**93**) (Howell & Martinez 1957), and the Elder Mt sandstone and Antrim plateau basalts of Western Australia (**94**, **95**) (Irving & Green 1958).

Pre-Cambrian (Figure 17). Creer (1957a) has sampled 12 sites in the Longmyndian (**96**). Samples from 111 sites spaced through 11 000 feet of the Torridonian Sandstone yield pole **98** for the Diabaig and lowest beds of the Applecross group, and pole **97** for the remainder of the Applecross and Aultbea groups (Irving & Runcorn 1957). Du Bois (1957) has obtained 5 pole positions from successive levels in the Middle and Upper Keeweenawan; 31 lava specimens from the Portage lake group give pole **99**, 25 specimens from red sandstones and lavas in the Copper Harbour pole **100**, 68 specimens from red sediments in the Freda and Nonesuch

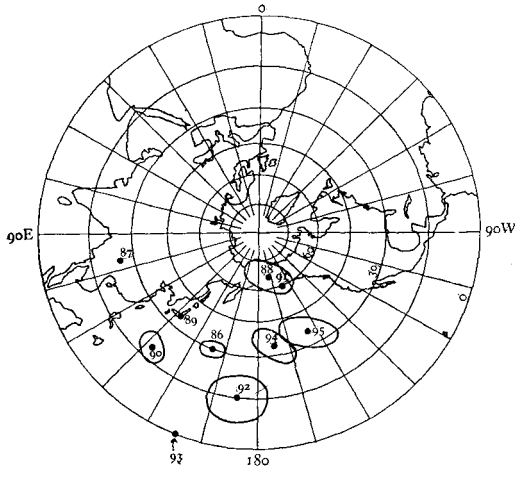


FIG. 16.—*Pole position for Cambrian, Silurian and Devonian.*

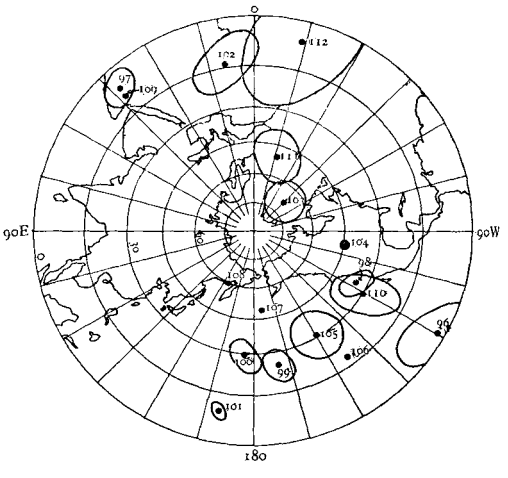


FIG. 17.—*Pre-Cambrian pole positions.*

group pole **101**, 15 specimens from the Jacobsville group pole **102**, and pole **103** has been obtained from 15 specimens in the Chequamegon group. Graham (1953) has measured the magnetization directions in 5 dykes from Baraba County, Michigan (**104**). Runcorn (1955) and Doell (1955) have given the pole for the Hakatai Shales of the Grand Canyon (**105**, **106**). Howell & Martinez (1957) have studied the red beds of the Hazel formation of west Texas and obtained pole **107** for flat-lying beds (5 sites) and **108** for folded beds (9 sites). The Pilansberg dykes measured by Gough (1956) (**109**) have recently been dated by isotope decay methods (Schreiner 1958). Three results, **110**, **111**, **112**, are available from Australia (Irving & Green 1958).

5. Discussion of results

For Devonian and earlier periods results are comparatively few, the time spans vast, and the results highly scattered. For Carboniferous and younger periods the results are numerous and the following remarks may be made about them.

(1) The results from the Neogene and Quaternary from different continents are in good agreement with the present geographic pole (Figures 8, 9 and 10). The

13652465b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

small divergences that do occur could be due to sampling inadequacies and small rotations of limited areas in tectonically active regions such as Japan. The Earth's field has remained predominantly an axial dipole.

(2) With a few exceptions (**36**, **44**) polar results from older rock formations diverge considerably from the present geographic pole. Polar results from Carboniferous and Permian rock formations are generally found to be in intermediate latitudes. Thereafter, in the Mesozoic and Palaeogene, they occupy higher latitudes.

(3) In *most* cases there is good but not always exact agreement between results from rock formations laid down in the *same* continent during the *same* geological period. The agreement between the ten results (**54** to **63**) from Permian beds in five European countries is most striking (Figure 14). Other examples are the Palaeogene of Europe (**18** to **23**, Figure 11), the lower sections of the Deccan Traps (**24** to **29** and **31**, Figure 11), the Jurassic of Africa (**41** to **43**, Figure 12), and the Triassic of the United States (**48** to **51**, Figure 13). Divergences do occur, for instance, pole **57**, obtained from measurements on a single lava flow from the Esterel, differs somewhat from the main group of determinations from the European Permian (Figure 14). However an adequate time average has not been achieved and the divergence is probably due to secular variation.

(4) The agreement referred to occurs between igneous and sedimentary rocks in a variety of geological environments.

(5) Polar results derived from rock formations laid down prior to the Neogene and in the *same* geological period but situated in *different* continents in general do not agree with one another.

(6) Positive and negative polarizations are found in all periods except in the Permian, in which only positive polarizations have so far been recorded, and in the Ordovician, for which there are no results.

(7) The polar results from various formations in the Carboniferous of the United Kingdom show wide scatter (Figure 15).

6. Comparison of results within and between continents

Of the points listed above the contrast in Permian to Palaeogene times of agreement between results from the *same* continent and disagreement between results from *different* continents is of first importance. The remainder of this review is devoted to a discussion of this matter. An informative procedure for comparing results within and between continents is to plot frequency histograms of the separation of poles taken two at a time first from the same and then from different continents. Poles must be used from the same unit of geological time, the unit being chosen sufficiently long to allow correlation of the unit as a whole from continent to continent, that is, a geological period for the time considered. For the internal comparison if n poles have been calculated from such a time unit within one continent there are $\frac{1}{2}n(n-1)$ pole pairs. For the external comparison, if $n_1, n_2, \ldots, n_m$ poles are known from m continents, the number of pole pairs is $\frac{1}{2}\{(\Sigma n)^2 - \Sigma n^2\}$. In the first case an estimate is obtained of the amount of polar wandering as viewed from one continent during a known period of geological time. This should agree with the external comparison if polar wandering *only* has occurred.

This analysis has been done for seven time intervals all of approximately equal duration. They are the Quaternary and Neogene approximately 28 million years, the Palaeogene 32 million years, Jurassic 25 million years, Triassic 30 million

13652465b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

years, Permian 25 million years, the Upper Carboniferous 25 million years and Lower Carboniferous 30 million years. All the results from the Deccan Traps are included in the analysis for the Palaeogene although this formation may extend down into the uppermost Cretaceous. The results for the Quarternary and Neogene are given in Figure 18. Both internal and external agreement is good, the divergences commonly ranging up to 20°. The Permian results, which are typical of the earlier periods, are plotted in Figure 19 and the results for the Permian, Triassic, Jurassic and Palaeogene combined in Figure 20. The divergence between pairs of poles within each time unit from a single continent range up to 30° with very few higher values. This is about 10° more than the range for the

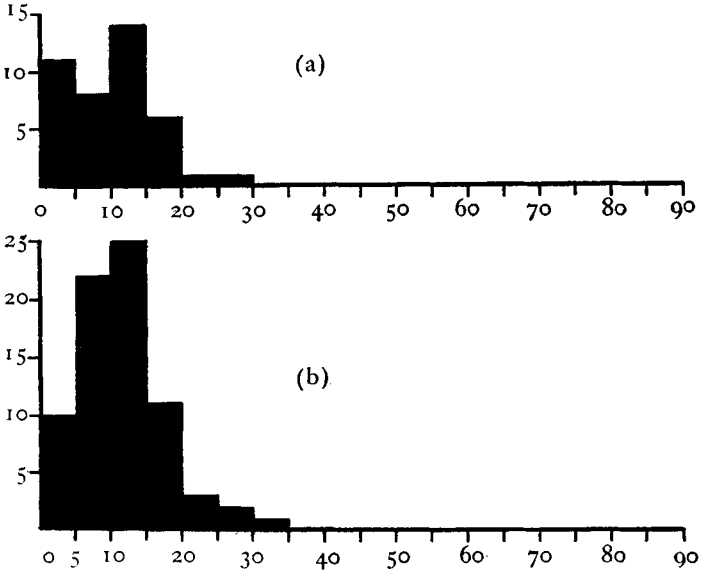


FIG. 18.—*Variation in the Quaternary and Neogene.* The height of each column in the histograms in Figures 18 to 21 represents the number of occurrences of divergences of pole pairs in the ranges indicated on the horizontal axis.

(a) within continents (b) between continents.

Quaternary and Neogene. The sampling from the latter is concentrated in the Quaternary (12 poles) with fewer determinations (4 poles) for the Neogene which represents a much longer period. Thus, the range shown in Figure 18 may be too small, and 30° could represent a more accurate range of polar wandering during periods of approximately 20 to 40 million years. In contrast, the pairs for different continents have separations commonly ranging between 35° and 90° and rarely approach closer than 20°.

In the Carboniferous the scatter of poles in the European results is greater than that for other epochs. The Carboniferous period lasted for approximately 55 million years, or about twice as long as the younger units, and for this reason the results are divided into Upper and Lower groups corresponding approximately to the Pennsylvanian and Mississippian (25 and 30 million years). The external divergences (Figure 21) are again greater than the internal although the latter are more scattered than in later periods. Thus, it seems that in the Carboniferous polar movement may have been greater than that in subsequent periods. The

13652465b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

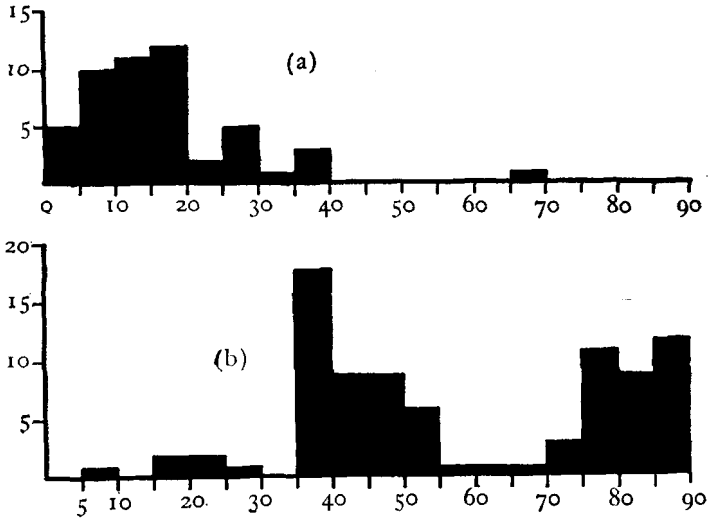


FIG. 19.—*Variation in the Permian.*
(a) within continents (b) between continents.

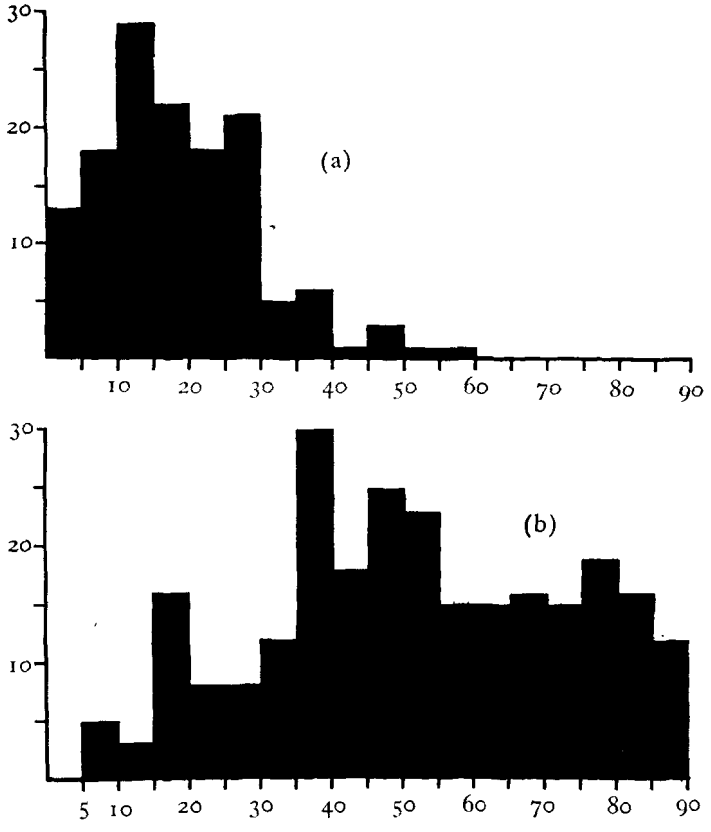


FIG. 20.—*Combined results from the Permian, Triassic, Jurassic and Palaeogene.*
(a) within continents (b) between continents.

13652465b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

results are not numerous and a reliable estimate cannot be made. It is interesting to note that in Australia between Silurian and Devonian (poles **91** and **88**) and Upper Carboniferous (**76**, **77**) there is a polar change not less than 60° of arc accomplished during the period covered by the results from the United Kingdom.

In pre-Carboniferous periods the picture is less clear because of the limited sampling over very much longer time intervals. Before analysis of the type described above can be made it will be necessary to obtain results from each epoch within each period, since epochs defined by geologists for Devonian and older systems are comparable in length to Mississippian and later periods.

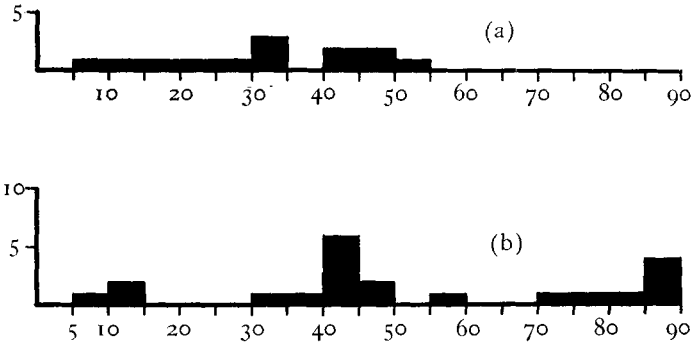


FIG. 21.—*Variation in the Carboniferous*
(a) within continents (b) between continents.

This contrast in the variation within and between continents illustrated in Figures 18 to 20 may be explained in terms of the failure of rocks to "fossilize" the geomagnetic field, variations in the behaviour of the field, inadequate stratigraphic correlation between continents, tectonic movements within continents, or relative displacements of the continents.

Failure of rocks to preserve the direction of the geomagnetic field. Elongated or discoid ferromagnetic particles in sediments are likely to be magnetized along their longer axes. Upon settling they will tend to lie horizontally rather than along the magnetic field so that the resultant inclination is too low. Compaction during lithification of a sediment could have a similar effect. Such effects do not apply to igneous rocks, or chemically magnetized sediments in which the ferromagnetic particles are of irregular shape. In rocks possessing a detrital magnetization it seems unlikely that important errors arise from this source. For instance, it is not present in the Torridonian Sandstone, as is shown by the parallel directions of magnetization in slumps and enclosing undisturbed beds (Irving 1957). Also, in varved sediments, which are the other rocks whose magnetization is attributed to detrital grains, the mean direction is parallel to the dipole field (Figure 8, **3** and **10**) and the inclination error observed (Johnson & others 1948, King 1955) in redeposition experiments in the laboratory either did not occur during deposition under natural condition or has been corrected since.

In aeolian and water-laid sediments elongated or discoid ferromagnetic particles magnetized along their long axis could have their alignment affected by wind or water currents. No aeolian beds have yet been studied so such an effect is relevant only to water-laid sediments possessing a detrital magnetization. The

13652465b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

agreement between igneous and sedimentary rocks discounts the effect. A detailed discussion has been given by Irving (1957).

Secondary I.R.M. components are common and may obliterate the primary magnetization. In the results from pre-Miocene rocks considered here this has not happened since the directions lie far from the present field. However, less obvious secondary components may be present so that the observed directions lie in the plane containing the present field and the primary component (Figure 7). In rock formations containing positive and negative polarizations in approximately equal proportions the average direction irrespective of sign is that of the primary magnetization, the secondary component being lost so that the effect only applies to those rock formation in which one polarity is present or in which the two polarities are disproportionate. Runcorn (1956a) shows that the effect is to change the *latitude* only of the calculated pole; in rock formations with negative polarization the calculated pole is too northerly, and in the case of positive polarization the pole is too far south. For instance, the Triassic sandstones of France have positive polarization somewhat affected by a secondary component of magnetization (Clegg & others 1957). The pole is to the south of, but in the same longitude as, the pole calculated from the Triassic sandstones of England (Clegg & others 1954) which have both positive and negative polarities. Secondary components cannot, however, account for the wide longitudinal spread of polar results.

The argument that palaeomagnetic results allow the behaviour of the Earth's field in the remote geological past to be determined has met with much criticism because of the possibility that the stress history of rocks could cause permanent deflections of the primary magnetization (Graham 1956, Graham & others 1957). Two situations are envisaged in which this may occur: (1) in both igneous and sedimentary rocks as a result of stresses operative during tectonic activity, and (2) in igneous rocks which acquire a T.R.M. in the direction of the Earth's field under stresses which are subsequently relieved. For example, hyperbyssal rocks become magnetized under non-hydrostatic load, and in these and in lava flows, cooling stresses may be operative at the time the T.R.M. is acquired. Relaxation occurs by jointing, by removal of the overburden through erosion, or by removal of the specimen from the outcrop.

Evidence for the first of these has been put forward by Graham & others (1957) who have shown in the laboratory that linear compressions of 250 kg/cm^2 can change the directions of magnetization in certain metamorphic rocks by as much as 30°. This is presumptive evidence only since the deflections are not permanent; with few exceptions the magnetizations resume their original orientations after the stress is released. In any case the material used is irrelevant. The specimens tested are Pre-Cambrian gneisses which are not useful for geomagnetic investigations for obvious reasons. In the experiments deflections occurred only in magnetite or ilmenomagnetite-bearing gneisses whereas the directions of magnetization in gneisses containing haematite as their chief ferromagnetic constituent were little affected; in nature the agreement between lavas containing magnetite and red haematitic sediments of comparable age is good (poles **54** to **63** poles **48** to **51**, pole **100**) and such magnetostrictive differences are not apparent. Also, where secondary components are unimportant, folded and unfolded beds from the same period and continent have consistent directions of magnetization and compressive forces sufficient in some cases to produce vertical tilt have not affected the directions of magnetization (Graham 1949, Irving 1957).

The second case is rendered most improbable by the good agreement observed

13652465b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

between the directions of magnetization in igneous and sedimentary rocks of the same geological age (see above), and by the laboratory experiments of Stott and Stacey (1959). They have shown that the directions of T.R.M. acquired by igneous rocks under directed pressure of 500 kg/cm^2 which is later released is parallel to the Earth's field in the laboratory and to the direction in duplicate specimens magnetized without stress. The specimens used are relevant material from lava flows and sills in Australia from which inferences regarding the direction of the ancient Earth's field have already been made. It has to be remembered however, that magnetic minerals often have their properties altered by heat treatment, and the properties of rock specimens after heating above the Curie point are not necessarily identical with those possessed by the rock initially. The extrapolation of results from heating experiments to actual situations in nature is not always a very certain matter.

Nevertheless, assuming magnetostrictive effects to be important in controlling the direction of magnetization of rocks it may be proposed that the Earth's elementary dipole has not changed its attitude throughout geological time and the observed polar successions are the result of varying stress histories. These would need to have been constant over continental areas but radically different in each continent for given periods of time. There is no geological basis for such a contention especially since the rock formations studied have been chosen because of their unaltered character and the absence of dynamic metamorphism. Such stress effects if present are likely to be too subtle to be detected geologically. A relevant test of such a hypothesis is to consider the consistency between results for a given period from a variety of geological environments in the same continent. If the range of stress histories found in these environments have important effects the results should not agree. There are uniform results from the Triassic of America (Du Bois & others, 1957, Figure 13) and the Permian of Europe (Figure 14), both including sediments and igneous rocks, some folded and others horizontal. Vertical magnetizations (that is directions perpendicular to the bedding) are found in a wide range of geological environments in Australia; Mesozoic dolerite intrusives magnetized under load (**45**), flat lying Permian lavas (**70**), and folded Carboniferous sediments and lavas (**76, 77**). It is clear that there is no substantial basis in the observations available at present from either the field or laboratory for supposing that magnetostrictive effects are responsible for the contrast in the observed variation within and between continents.

Variations in the geomagnetic field. This contrast could result from variations in the behaviour of the field. There are two possibilities, either that the field has not remained a dipole field throughout geological time, or that it has remained a dipole field which, prior to Miocene times, underwent rapid oscillations of large amplitude, so that during the space of a geological period the pole traversed much of the Earth's surface. At first sight it would seem that the agreement between results for the same period from the same continent effectively discounts the latter. However, because of the possibilities of gross inadequacies in stratigraphic correlation and possible widespread thermal and chemical changes this is not necessarily so.

Inadequate stratigraphic correlation. In any continent the rock formations of a geological period represent only a proportion of the total time span of that period. In general, sections in different continents overlap, but the fractions of these which have received palaeomagnetic study probably do not, so that the record in different continents will deal with different fractions of time within each geological period.

13652465b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

If, say, within Permian times polar movements with amplitudes of 90° and periods of tens of millions of years occurred, the pole could appear to be approximately stationary in the time sample from one continent, whereas the longer time span covered by specimens from several continents could contain a wide scatter of poles, as is observed. Such polar oscillations must have decreased some time in the middle of the Tertiary, after which they became negligible (Figures 9 and 10). Now, for post-Carboniferous times, polar shifts between geological periods in the same continent are small. For instance, during the period from Upper Carboniferous to Jurassic the pole as measured from Australia has remained in the region of the Tasman Sea. Therefore the large polar oscillations postulated above must have followed approximately repeatable paths. Moreover they were in phase with the processes by which rocks became magnetized; that is, rock formations in successive geological periods in the same continent became magnetized when the pole was moving through similar sections of its cycle. These sections are different for different continents for the same geological period, being about 90° of arc away from one another in the case of Europe and Australia. Any such connection between polar oscillation and the matters referred to in section 2 seems unlikely. A further possibility is that the system boundaries are grossly diachronous. The observed difference of within- and between-continent variation could arise if the systems in different continents were out of step by twice or three times their length. This is geologically absurd for the Permian and later periods here considered.

Continent-wide heating and chemical effects. Let us suppose again a widely oscillating dipole field and that rocks become magnetized long after deposition by heating above the Curie temperature during burial, or, by chemical changes affecting the iron minerals. The contrast in the within and between continent variation could then be explained if these processes were in phase with the polar oscillations. This is unlikely, and in any case the chemical and heat treatment of the rock formation of the same geological period from different parts of the same continent would have to have been identical. The variety of structure within a continent makes this improbable. Furthermore heating above the Curie temperature for red sediments (Curie temperature 600 to 700°C) implies burial to 20 km or more which is not likely to have happened to the rock formations considered here. Lastly, such widespread chemical changes would almost certainly be visible petrologically. For instance, certain sandy dolomites from the Sawatch formation of Cambrian age from Colorado have directions of magnetization parallel to those observed in Carboniferous rocks from the same region (Howell & Martinez 1957). Petrological study suggests that these sediments were dolomitized long after deposition and this was accompanied by the partial replacement of glauconite by haematite, so that a magnetization appropriate to a later period is not surprising in this case. Substantial secondary chemical changes of this type are not apparent in the rock formations listed in the table which were chosen for study because of their unaltered character.

Tectonic movements within a continent. Relative rotations of regions within a continent would result in differing pole results, and the position of the Eocene pole (**35**) from Oregon could be explained in this way. The better agreement of poles within a continent compared with that between continents is evidence against the occurrence of such rotations, unless these have contrived to bring previously dispersed poles into agreement. It seems much more likely that tectonic rotations would increase dispersion.

13652465b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The assumption of a geocentric dipole field. The disagreement of results from different continents could be evidence against the assumption of a geocentric dipole field. However, there are arguments in favour of such a field.

Continued westerly drift of the Earth's field at the present rate would result in an average dipole field in about 2 000 years. The westerly drift is attributed to the acceleration of the mantle relative to the core in which the field is generated. Thus, whatever the field in the core, it would appear on the surface as a dipole when averaged over periods of several thousands of years provided no substantial part of the field arises in the mantle (Creer & others 1957). This is consistent with data from lavas of Mt Etna and the Swedish varves which have been sampled for 2 305 and 1 850 years respectively and which average to the geocentric dipole field (Figure 8).

Further back in time the field as observed from four continents over the past million years has averaged to a dipole (Figure 9) and Neogene observations going back approximately 20 million years from Europe and North America give a similar result (Figure 10). Still further back in time, the internal consistency of results from the Triassic of the United States and the Permian of Europe indicate that for these periods of time the field has averaged to a dipole over areas of continental size. Moreover, it may be said with less precision, that the field for Permian and Triassic period as viewed from *both* Europe and North America has averaged approximately to a dipole as the poles all lie in the region of East Asia and the North Western Pacific. Statistical differences separate the poles from the two areas but the predominant component is dipolar.

Serial reversals of magnetization in Quaternary and Neogene rock formations indicate several reversals of the field during this time. These are thought to be intimately bound up with the average dipole nature of the field. Reversals of magnetization observed in rocks of greater age are not different in character (Irving & Runcorn 1957) and their occurrence is thus consistent with an average dipole field during previous periods.

Runcorn (1954) has argued that the Coriolis force dominates fluid motions in the core leading to the coupling of the geomagnetic dipole axis and the rotational axis. This is certainly true for the past 20 million years and if it has always been so the pole obtained from palaeomagnetism may be identified with the ancient geographic pole, which in turn may be checked against the palaeoclimatic evidence collected by geologists. It is found, in general, that rock formations containing evidence of hot climates have low magnetic inclinations whereas those supposed by geologists to have been deposited in cold climates have steep inclinations (Runcorn 1956b, Irving 1956a, Irving & Green 1958). This is consistent with an average dipole field parallel to the axis of rotation.

Thus, the hypothesis of an average dipole field is an extrapolation into remote geological time of the known behaviour of the field over the past 20 million years, and is analogous to the uniformitarian principles generally used in geology. Without this hypothesis the palaeomagnetic data imply that prior to the Miocene the field was not predominantly dipolar. This field was quite different from the non-dipole field nowadays since it remained approximately constant for periods of the order of 20–30 million years but underwent slow changes over periods of the order of hundreds of millions of years.

Relative continental displacement. The good agreement of poles within a continent but poor agreement between continents could mean that the continents behaving

13652465b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

as separate entities have moved relative to one another prior to the middle Tertiary. One possibility is that of an expanding earth with continents carried out along extending radii and no change in latitude or longitude. But this could not explain the variation between continents since the polar calculations are independent of the Earth's radius (Creer & others 1957). The second possibility is that of relative changes in longitude or rotations of the continents. Such movements can adequately explain the variation between continents. Attempts have been made to calculate these movements from the palaeomagnetic data (Du Bois 1957, Creer & others 1958, Irving 1958) and will not be repeated here. These are found to be broadly consistent with the continental displacements already suggested by Wegener (1924), Du Toit (1937), Carey (1955) and King (1953) on the basis of morphological, geological and biological evidence. The movements are predominantly longitudinal because the differences between poles for the same period for different continents are predominantly longitudinal, and are of the order of 10 000 km.

7. Concluding remarks

The study of rock magnetism is in its infancy, and is perhaps at about the same stage as palaeontology a century or more ago. Palaeontology progressed by systematic studies of faunas and floras, and in a similar way rock magnetism is likely, for the next decade at least, to benefit most from systematic descriptions of the magnetic properties of rock formations—their directions of magnetization, the stability of these directions studied by field and laboratory tests, and the properties of their ferromagnetic minerals. The reconnaissance stage of palaeomagnetism, when small numbers of specimens were taken from many rock formations, is closed, and detailed studies of single stratigraphic entities are required. The interpretation of palaeomagnetic data presents many difficulties and, in the words of Graham & others (1957) "the only obvious way of these is to have so many field observations from so many rocks of so many types in so many settings that the insidious influence of magnetostriction either taken alone or in conjunction with (other?) time dependent parameters is eliminated". It is gratifying to see that so much of this type of information is now coming to hand.

Acknowledgments

I am indebted to Professor J. C. Jaeger, Dr F. D. Stacey and Mr R. Green of this University for much help in discussion.

Australian National University,
Canberra, A.C.T.,
Australia.
1958 *December.*

References

Allan, D. W., 1958. *Nature* (*London*), **182,** 469.
Asami, E., 1954. *Proc. Japan. Acad.*, **30,** 102.
Belshé, J. C., 1957. *Adv. in Phys.*, **6,** 187.
Blundell, D. J., 1957. *Geol. Mag.*, **94,** 187.
Bruckshaw, J. M. & Robertson, E. I., 1949. *Mon. Not. R. Astr. Soc. Geophys. Suppl.*, **5,** 308.

13652465b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Bruckshaw, J. M. & Vincenz, S. A., 1954. *Mon. Not. R. Astr. Soc. Geophys. Suppl.*, **6**, 579.
Brynjolfson, A., 1957. *Adv. in Phys.*, **6**, 247.
Bullard, E. C. & Gellman, H., 1954. *Phil. Trans. Roy. Soc. A*, **247**, 213.
Bullard, E. C., Freedman, C., Gellman, H. & Nixon, J., 1950. *Phil. Trans. Roy. Soc. A*, **243**, 67.
Campbell, C. D. & Runcorn, S. K., 1956. *J. Geophys. Res.*, **61**, 449.
Carey, S. W., 1955. *Pap. & Proc. Roy. Soc. Tasmania*, **89**, 255.
Chapman, S. & Bartels, J., 1951. *Geomagnetism*, **2**, 750 (Oxford).
Chevallier, R., 1925. *Ann. phys. Paris*, **4**, 5.
Clegg, J. A., Almond, M. & Stubbs, P. H. S., 1954. *Phil. Mag.*, **45**, 583.
Clegg, J. A., Deutsch, E. R. & Griffiths, D. H., 1956. *Phil. Mag.*, **1**, 419.
Clegg, J. A., Deutsch, E. R., Everitt, C. W. F. & Stubbs, P. H. S., 1957. *Adv. in Phys.*, **6**, 219.
Clegg, J. A., Radakrishnamurty, C. & Sahasrabudhe, P. W., 1958. *Nature (London)*, **181**, 830.
Coombs, D. G., 1958. *Austr. J. Sci.*, **21**, 18.
Cox, A. A., 1957. *Nature (London)*, **179**, 685.
Creer, K. M., 1957a. *Phil. Trans. Roy. Soc. A*, **250**, 111.
Creer, K. M., 1957b. *Phil. Trans. Roy. Soc. A*, **250**, 130.
Creer, K. M., 1958. *Ann. Geophys.*, **14**, 393.
Creer, K. M., Irving, E. & Runcorn, S. K., 1954. *J. Geomag. Geoelec.*, **6**, 163.
Creer, K. M., Irving, E. & Runcorn, S. K., 1957. *Phil. Trans. Roy. Soc. A*, **250**, 144.
Creer, K. M., Irving, E., Nairn, A. E. M. & Runcorn, S. K., 1958. *Ann. Geophys.*, (in press).
Deutsch, E. R., Radakrishnamurty, C. & Sahasrabudhe, P. W., 1958. *Phil. Mag.*, **3**, 170.
Doell, R. R., 1955. *Nature (London)*, **176**, 1167.
Doell, R. R., 1956. *Trans. Amer. Geophys. Un.*, **37**, 156.
Doell, R. R., 1957. *Adv. in Phys.*, **6**, 327.
Du Bois, P. M., 1957. *Adv. in Phys.*, **6**, 177.
Du Bois, P. M., Irving, E., Opdyke, N. D., Runcorn, S. K. & Banks, M. R., 1957. *Nature (London)*, **180**, 1186.
Du Toit, A. L., 1937. *Our Wandering Continents* (London, Oliver & Boyd).
Elsasser, W. M., 1956. *Rev. Mod. Phys.*, **28**, 135.
Gough, D. I., 1956. *Mon. Not. Roy. Astr. Soc. Geophys. Suppl.*, **7**, 196.
Graham, J. W., 1949. *J. Geophys. Res.*, **54**, 131.
Graham, J. W., 1953. *J. Geophys. Res.*, **58**, 243.
Graham, J. W., 1955. *J. Geophys. Res.*, **60**, 329.
Graham, J. W., 1956. *J. Geophys. Res.*, **61**, 735.
Graham, J. W., Buddington, A. F. & Balsley, J. R., 1957. *J. Geophys. Res.*, **62**, 465.
Graham, K. W. T. & Hales, A. L., 1957. *Adv. in Phys.*, **6**, 149.
Green, R., 1958. *Nature (London)*, **182**, 382.
Griffiths, D. H., 1955. *Mon. Not. R. Astr. Soc. Geophys. Suppl.*, **7**, 103.
Haigh, G., 1958. *Phil. Mag.*, **3**, 267.
Hatherton, T., 1953. *New Zealand J. Sci. Tech.*, **35**, 421.
Hide, R., 1957. Hydrodynamics of the Core, *Physics and Chemistry of the Earth*, (London: Pergamon Press), **2**, 94.

13652465b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Hospers, J., 1954. *Kon. Ned. Akad. Wet.*, B, **56,** 467.
Hospers, J., 1955. *J. Geol.*, **63,** 59.
Hospers, J. & Charlesworth, H. A. K., 1954. *Mon. Not. R. Astr. Soc. Geophys. Suppl.*, **7,** 32.
Howell, L. G. & Martinez, J. D., 1957. *Geophysics*, **22,** 384.
Howell, L. G., Martinez, J. D., & Statham, E. M., 1958. *Geophysics*, **23,** 285.
Irving, E., 1956a. *Geofisica pura e app.*, **33,** 23.
Irving, E., 1956b. *Pap. and Proc. Roy. Soc. Tasmania*, **90,** 157.
Irving, E., 1957. *Phil. Trans. Roy. Soc. A*, **250,** 100.
Irving, E., 1958. *Geophys. J.*, **1,** 224.
Irving, E., & Green, R., 1957. *Mon. Not. R. Astr. Soc. Geophys. Suppl.* **7,** 347.
Irving, E. & Green, R., 1958. *Geophys. J.*, **1,** 64.
Irving, E. & Runcorn, S. K., 1957. *Phil. Trans. Roy. Soc. A*, **250,** 83.
Ising, G., 1942. *Arkiv. Mat. Astr. Fysik*, **29a,** 42.
Johnson, A. E., Murphy, T. & Torreson, O. W., 1948. *Terr. Mag.*, **53,** 349.
King, L. C., 1953. *Bull. Amer. Assoc. Pet. Geol.*, **37,** 2163.
King, R. F., 1955. *Mon. Not. R. Astr. Soc. Geophys. Suppl.*, **7,** 115.
Krishnan, M. S., 1949. *Geology of India and Burma*, (Madras), page 431.
Leng, J. H., 1955. *Thesis*, University of London.
Martinez, J. D. & Howell, L. G., 1956. *Nature* (*London*), **178,** 204.
Mercanton, P. L., 1926. *Terr. Mag.*, **31,** 187.
Nairn, A. E. M., 1956. *Nature* (*London*), **178,** 935.
Nairn, A. E. M., 1957a. *Adv. in Phys.*, **6,** 162.
Nairn, A. E. M., 1957b. *Mon. Not. R. Astr. Soc. Geophys. Suppl.* **7,** 308.
Nairn, A. E. M., 1958. *Bull. Soc. Geol. France*, **7,** 721.
Nagata, T., 1953. *Rock Magnetism*, (Maruzen, Tokyo), page 179.
Nagata, T., Akimoto, S., Uyeda, S., Shimizu, Y., Ozima, M. & Kogayashi, K., 1957. *Adv. in Phys.*, **6,** 255.
Néel, L., 1951. *Ann. Geophys.*, **7,** 90.
Néel, L., 1955. *Adv. in Phys.*, **4,** 191.
Rimbert, F., 1958. *Thesis.* Paris.
Roche, A., 1950a. *C. R. Acad. Sci. Paris*, **230,** 113.
Roche, A., 1950b. *C.R. Acad. Sci. Paris*, **230,** 1603.
Roche, A., 1951. *C.R. Acad. Sci. Paris*, **233,** 1132.
Roche, A., 1957. *C.R. Acad. Sci. Paris*, **243,** 812.
Roche, A., Cattala, L. & Boulanger, J., 1958. *C.R. Acad. Sci. Paris*, **246,** 2922.
Runcorn, S. K., 1954. *Trans. Amer. Geophys. Un.*, **35,** 49.
Runcorn, S. K., 1955. *Bull. Geol. Soc. Amer.*, **67,** 301.
Runcorn, S. K., 1956a. *Geol. Assoc. Canada*, **8,** 77.
Runcorn, S. K., 1956b. *Geol. Mijnbouw*, **18,** 253.
Rutten, M. G., Everingen, R. O. & Zijderveld, J. D. A., 1957. *Geol. Mijnbouw*, **19,** 193.
Schreiner, G. D. L., 1958. *Nature* (*London*), **181,** 1330.
Sigurgeirsson, Th., 1957. *Adv. in Phys.*, **6,** 240.
Stott, P. M. & Stacey, F. D., 1959. *Nature* (*London*), **183** (in press).
Torreson, O. W., Murphy, T., & Graham, J. W., 1949. *J. Geophys. Res.*, **54,** 111.
Vestine, E. H., Laporte, L., Lange, I. & Scott, W. E., 1947. *Carnegie Institution Publications* Washington, No. 580.
Vincenz, S. A., 1954. *Mon. Not. R. Astr. Soc. Geophys. Suppl.*, **6,** 590.
Wegener, A., 1924. *Origin of Continents and Oceans*, (London, Methuen).

13652465b, 1959, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1959.tb05781.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License